

Executive Summary

The hospital analysis was performed on **1,000** patient records covering the period **1 January 2025 to 31 December 2025**.

The analysis examined patient demographics, departments, diseases, billing, insurance, length of stay, readmission and satisfaction.

The hospital generated approximately **₹7,65,06,096** in total billing, with an average bill of **₹76,506.10**. **Neurology** handled the highest number of patients (143), while Oncology generated the highest revenue (**₹1,45,15,225**).

The most frequently recorded disease was **Heart Disease**, with **111 cases**. The average length of stay was **8.07 days**, and the overall readmission rate was **16.6%**.

Based on the analysis, management should focus on resource allocation in high-patient departments, strengthening high-revenue departments, improving patient satisfaction, reducing readmissions, and monitoring healthcare costs.

BUSINESS INSIGHTS TEMPLATE

1. Department Volume

Finding: Neurology has the highest patient volume with 143 patients.

Business Meaning: This department may require more doctors, nurses, beds, and medical resources.

2. Revenue

Finding: Oncology generates the highest revenue of approximately ₹14.52 million.

Business Meaning: Management should monitor this major revenue driver and ensure sufficient treatment capacity.

3. Length of Stay vs Bill Amount

Finding: The correlation between Length of Stay and Bill Amount is 0.678.

Business Meaning: This shows a moderately strong positive relationship—longer hospital stays are generally associated with higher bills.

4. Patient Satisfaction

Finding: Orthopedics has the lowest average satisfaction score of approximately 3.96/5.

Business Meaning: Management should investigate waiting time, staff availability, and service quality.

5. Readmission Rate

Finding: Neurology has the highest readmission rate at approximately 20.98%.

Business Meaning: The department should improve discharge planning, patient monitoring, and follow-up procedures.

6. Most Common Disease

Finding: Heart Disease is the most common disease, with 111 patients.

Business Meaning: The hospital can allocate more specialists, equipment, and resources for cardiac care.

7. Age Group Analysis

Finding: The 66+ age group has the highest number of patients with 221 patients.

Business Meaning: Healthcare services should focus more on elderly patient care and age-related diseases.

8. Insurance Contribution

Finding: Self-Pay generates the highest revenue of approximately ₹28.20 million.

Business Meaning: The hospital should analyze payment patterns and improve financial planning for different payment categories.

9. Monthly Revenue

Finding: December has the highest hospital revenue at approximately ₹7.25 million.

Business Meaning: Management can analyze the reasons for increased revenue during December and plan resources accordingly.

10. High-Billing Patients

Finding: The average bill amount is approximately ₹76,506, while the highest bill reaches approximately ₹184,030. Also, 454 patients have bills above the average.

MANAGEMENT RECOMMENDATION

1. Increase Capacity in High-Volume Departments

Evidence: Neurology has the highest patient volume with 143 patients.

Recommendation: Increase doctors, nurses, beds, and equipment in Neurology to handle the higher workload.

2. Investigate Low Patient Satisfaction

Evidence: Orthopedics has the lowest average satisfaction score of approximately 3.96/5.

Recommendation: Analyze waiting time, staff availability, treatment quality, and patient service in Orthopedics.

3. Monitor Unusually Long Hospital Stays

Evidence: 329 patients stayed in the hospital for more than 10 days.

Recommendation: Review long-stay cases to identify treatment delays and opportunities to improve hospital efficiency.

4. Analyze Departments with High Readmission Rates

Evidence: Neurology has the highest readmission rate at approximately 20.98%.

Recommendation: Investigate the causes of readmission and improve treatment and discharge planning.

5. Improve Discharge and Follow-Up Processes

Evidence: High readmission rates indicate that some patients may require better post-discharge support.

Recommendation: Introduce follow-up calls, appointments, medication guidance, and patient monitoring after discharge.

6. Monitor High-Value Billing Cases

Evidence: The average hospital bill is approximately ₹76,506, while the maximum bill is ₹184,030. Also, 454 patients have bills above the average.

Recommendation: Analyze high-billing patients to understand expensive treatments and identify cost-management opportunities.

7. Improve Insurance and Payment Processing

Evidence: Self-Pay generates the highest revenue, approximately ₹28.20 million.

Recommendation: Improve payment collection processes and analyze insurance partnerships to maintain stable revenue.

8. Use Monthly Admission Trends for Staff Planning

Evidence: November has the highest number of admissions with 93 patients.

Recommendation: Plan additional doctors, nurses, beds, and resources during high-admission periods.

9. Monitor Doctor Workload

Evidence: Doctor workload is not equally distributed; Dr. Patil handled 73 patients, while some doctors handled fewer patients.

Recommendation: Balance patient allocation among doctors to reduce workload pressure and improve service quality.

10. Build a Recurring Hospital Analytics Dashboard

Evidence: The hospital dataset contains important KPIs such as patient volume, revenue, bill amount, satisfaction, length of stay, and readmission.

Recommendation: Build a regular Power BI dashboard to monitor these KPIs and help management make faster data-driven decisions.